(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

III rd Semester, November, 2022

Paper Code: CIC 205 **Subject: Discrete Mathematics**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Write the Principle of Inclusion and Exclusion ? Explain in brief.	2	CO1
(b)	Analyse Divide and conquer technique? Explain any one Example of it ?	2	CO2
(c)	Find the Converse and contrapositive of the Statement "If x is positive then $x \neq 0$"	2	CO1
(d)	Represent the statement using predicate and quantifier and negate it For all the real number x if $x > 5$ then $x^2 > 25$	2	CO1
(e)	Define lattices ?	2	CO2
2 (a)	Show that $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ is tautology By Rules of Preposition.	5	CO1
(b)	Prove that "if x,y ∈ Z (set of integer)such that xy is odd then both x and y are odd, by proving its contrapositive	5	CO1
3 (a)	Minimize the given function using K-map $f(x,y,z) = xyz + \bar{x}yz + x\bar{y}z + xy\bar{z} + \overline{xyz}$	5	CO2
(b)	What are the different solution methods of homogenous and non homogenous recurrence relation Explain with example?	5	CO2
4 (a)	(a) Let A ={1,2,3,4,6} and R is a Relation on the Set A Such that aRb if a divides b Find (i) Relation R (ii) Diagraph of R (iii) Find Adjacency matrix of R (iv) Indergree and outdegree of each node (v) Find its Hasse diagram	5	CO1
(b)	Analyse all the three cases of master method in solving Recurrance Relation	5	CO2